

1.Geospatial Data Visualization

Code:

```
install.packages("ggplot2")

library(ggplot2)

city <- data.frame(
  City=c("Chennai","Bangalore","Hyderabad","Mumbai","Delhi",
        "Pune","Kolkata","Ahmedabad","Jaipur","Kochi"),
  Latitude=c(13.08,12.97,17.38,19.07,28.61,18.52,22.57,23.02,26.91,9.93),
  Longitude=c(80.27,77.59,78.48,72.87,77.21,73.85,88.36,72.57,75.79,76.26),
  Population=c(11,13,10,20,32,7,15,8,4,2)
)

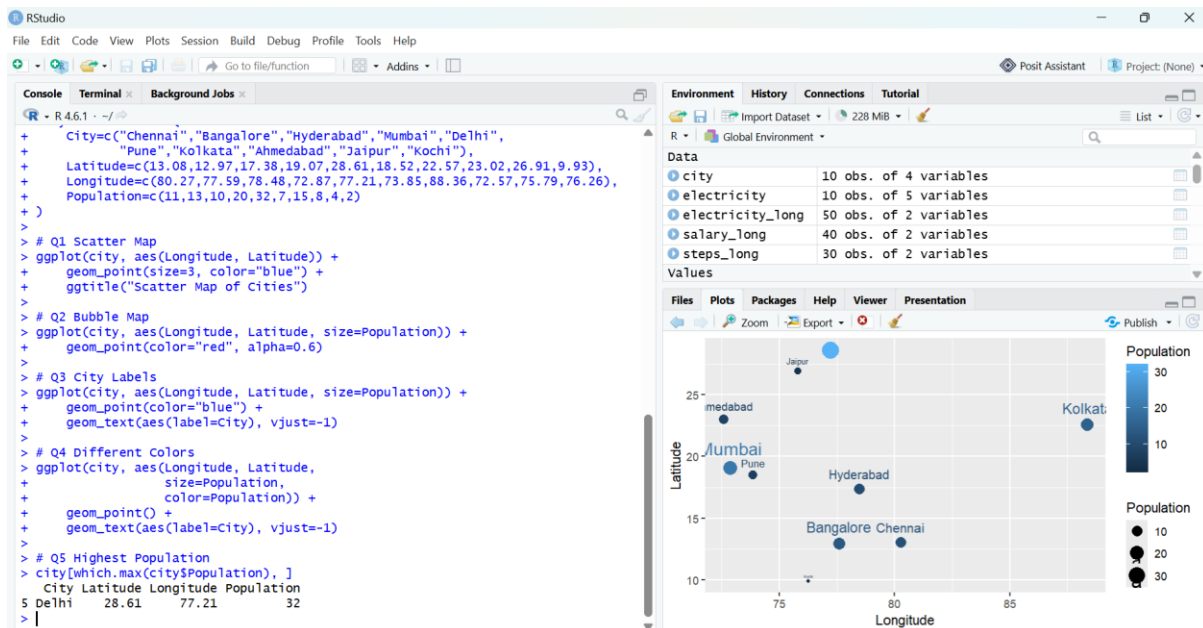
ggplot(city, aes(Longitude, Latitude)) +
  geom_point(size=3, color="blue") +
  ggtitle("Scatter Map of Cities")

ggplot(city, aes(Longitude, Latitude, size=Population)) +
  geom_point(color="red", alpha=0.6)

ggplot(city, aes(Longitude, Latitude, size=Population)) +
  geom_point(color="blue") +
  geom_text(aes(label=City), vjust=-1)

ggplot(city, aes(Longitude, Latitude,
                 size=Population,
                 color=Population)) +
  geom_point() +
  geom_text(aes(label=City), vjust=-1)

city[which.max(city$Population), ]
```



2. Network Visualization

Code:

```
install.packages("igraph")
```

```
library(igraph)
```

```
edges <- data.frame(
  From=c("A","A","B","C","C","D","E","F","G","H"),
  To=c("B","C","D","D","E","E","F","G","H","A")
)
```

```
g <- graph_from_data_frame(edges)
```

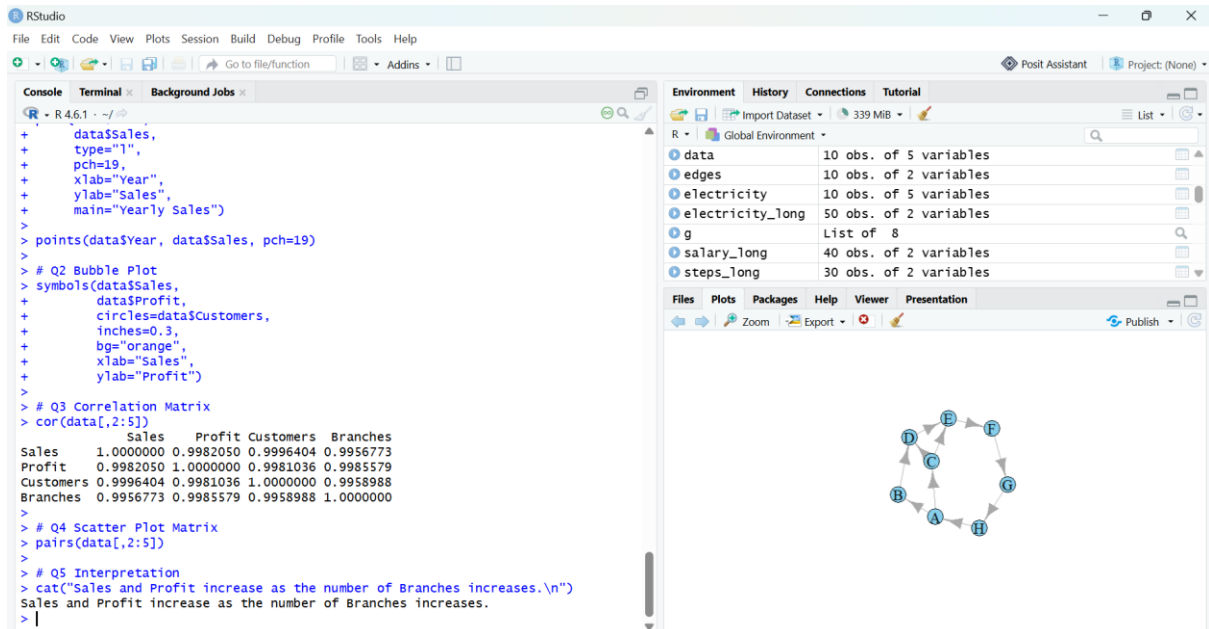
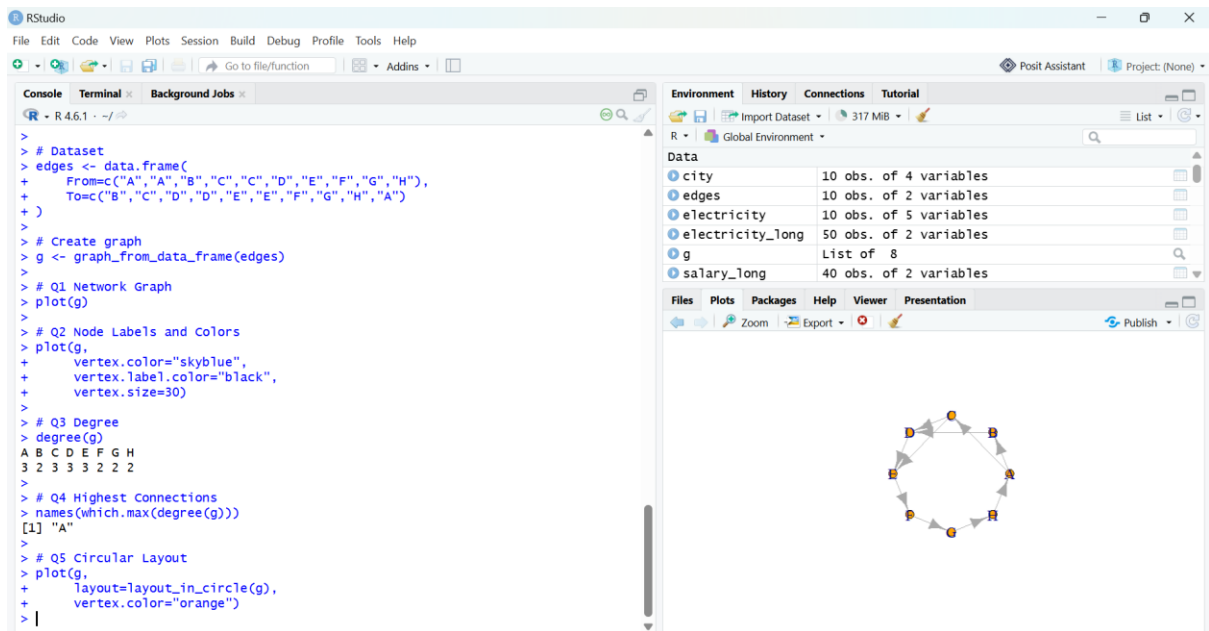
```
plot(g)
```

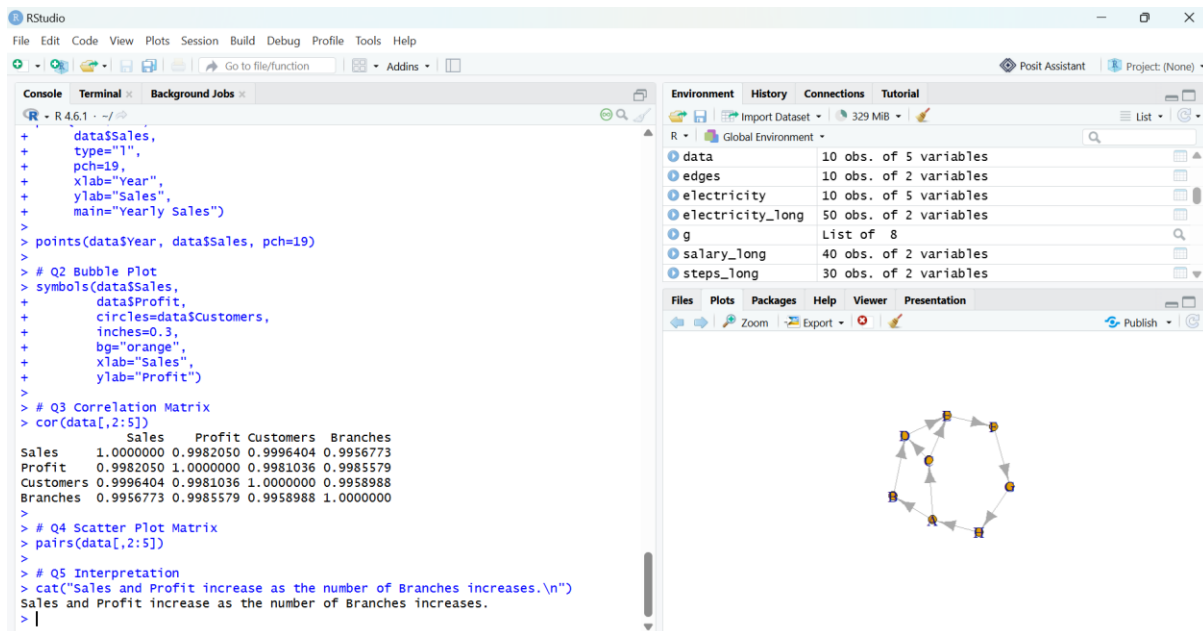
```
plot(g,
  vertex.color="skyblue",
  vertex.label.color="black",
  vertex.size=30)
```

```
degree(g)
```

```
names(which.max(degree(g)))
```

```
plot(g,
  layout=layout_in_circle(g),
  vertex.color="orange")
```





3. Temporal Data Representation

Code:

```
sales <- c(120,135,150,170,190,220,210,230,250,270,290,320)
```

```
months <- c("Jan", "Feb", "Mar", "Apr", "May", "Jun",
```

```
            "Jul", "Aug", "Sep", "Oct", "Nov", "Dec")
```

```
plot(sales,
```

```
      type="l",
```

```
      xaxt="n",
```

```
      xlab="Month",
```

```
      ylab="Sales",
```

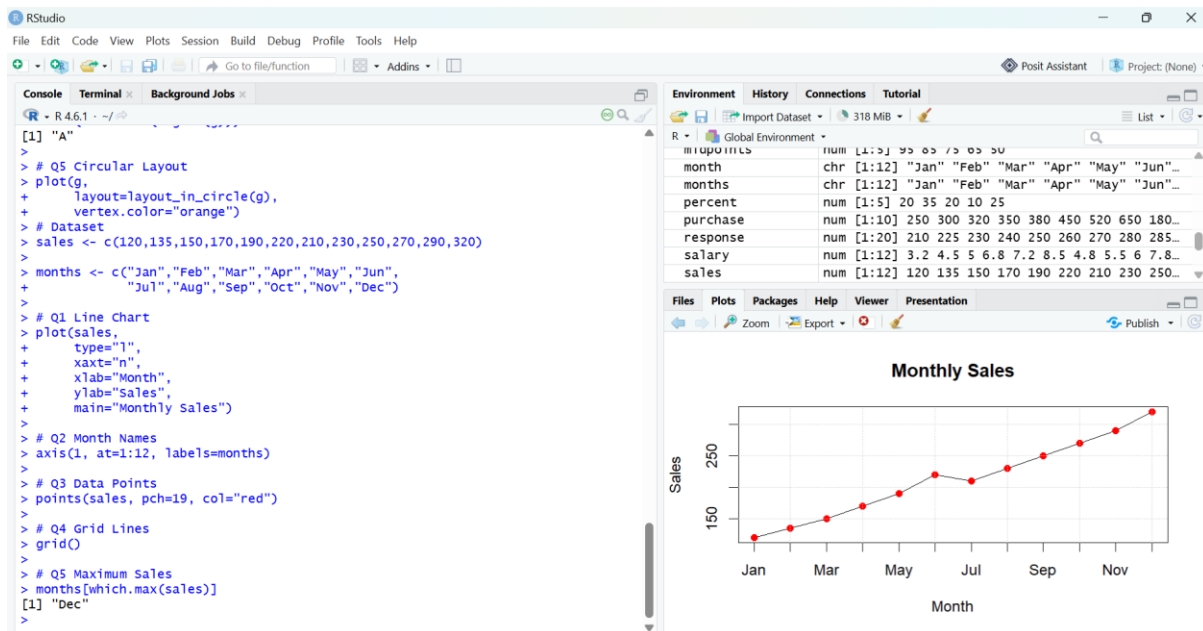
```
      main="Monthly Sales")
```

```
axis(1, at=1:12, labels=months)
```

```
points(sales, pch=19, col="red")
```

```
grid()
```

```
months[which.max(sales)]
```



4. Multivariate Data Visualization

Code:

```
student <- data.frame(
  Student=c("A","B","C","D","E","F","G","H","I","J"),
  Marks=c(60,68,74,79,82,85,88,91,94,97),
  Attendance=c(75,80,84,88,90,92,94,95,97,99),
  StudyHours=c(2,3,4,5,5,6,6,7,8,9),
  Projects=c(1,2,2,3,3,4,4,5,5,6))

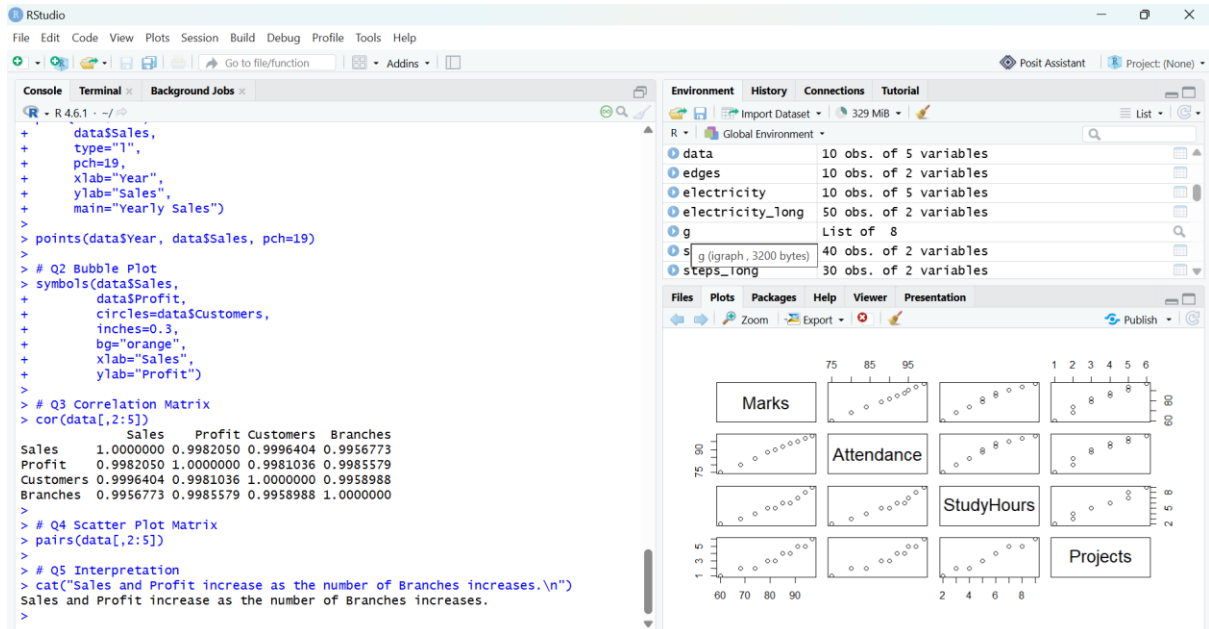
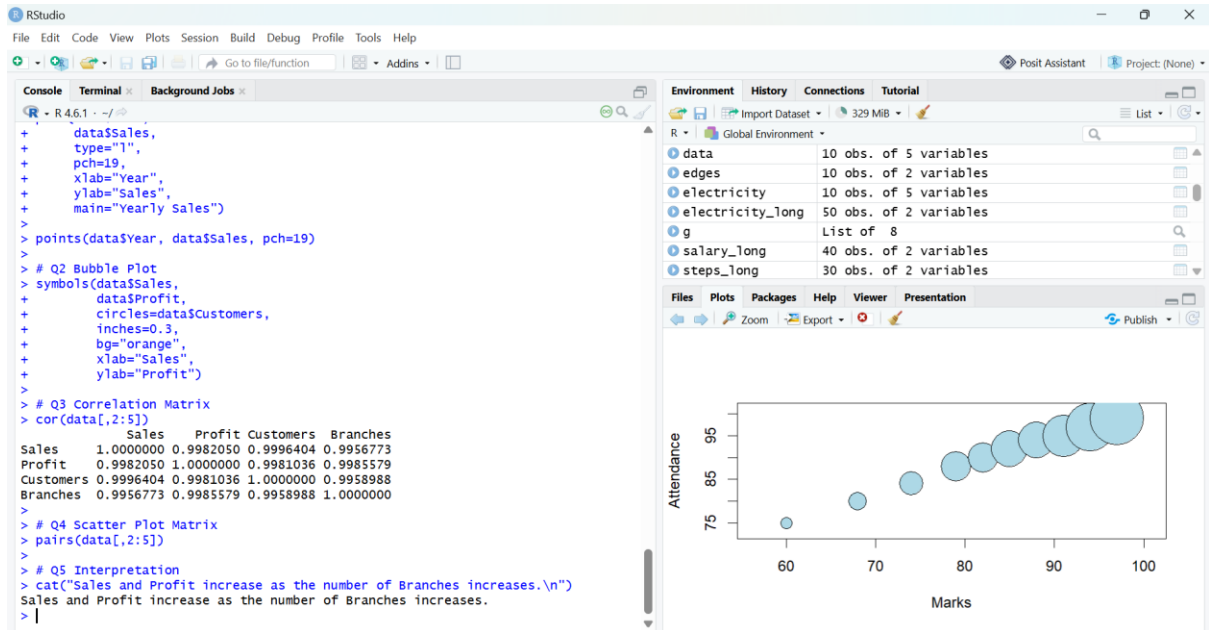
symbols(student$Marks,
        student$Attendance,
        circles=student$StudyHours,
        inches=0.3,
        bg="lightblue",
        xlab="Marks",
        ylab="Attendance")

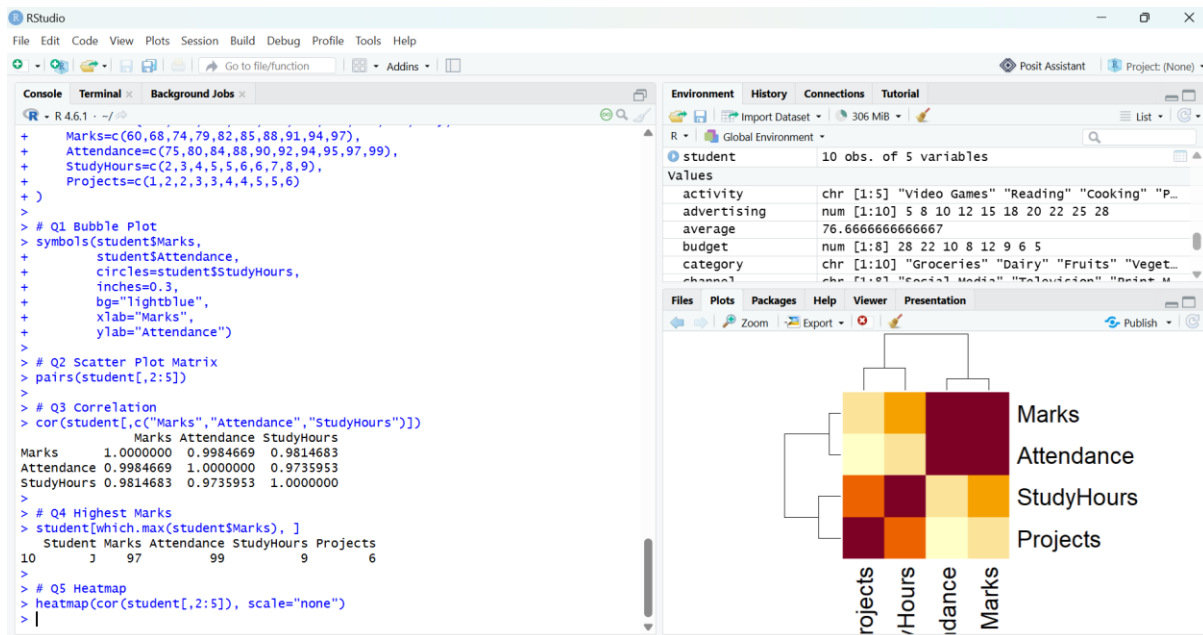
pairs(student[,2:5])

cor(student[,c("Marks","Attendance","StudyHours")])

student[which.max(student$Marks), ]

heatmap(cor(student[,2:5]), scale="none")
```





5. Integrated Data Visualization Exercise

Code:

```

data <- data.frame(
  Year=2016:2025,
  Sales=c(150,170,200,240,260,300,340,390,430,480),
  Profit=c(20,25,32,40,42,55,65,72,82,95),
  Customers=c(1200,1500,1800,2200,2400,2800,3300,3800,4200,4700),
  Branches=c(5,6,7,8,8,10,11,12,13,15)
)

plot(data$Year,
      data$Sales,
      type="l",
      pch=19,
      xlab="Year",
      ylab="Sales",
      main="Yearly Sales")

points(data$Year, data$Sales, pch=19)

symbols(data$Sales,
        data$Profit,

```

```

circles=data$Customers,

inches=0.3,

bg="orange",

xlab="Sales",

ylab="Profit")

cor(data[,2:5])

pairs(data[,2:5])

cat("Sales and Profit increase as the number of Branches increases.\n")

```

